

SAI KUMAR MEDIBOINA

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OBJECTIVE

Backend engineer specializing in performance optimization and distributed systems at scale, with a track record of cutting latency by up to 97% on enterprise workloads. Experienced in semantic search, hybrid retrieval, AI-assisted backend workflows, and high-throughput enterprise compliance platforms.

EXPERIENCE

Software Application Engineer

Oracle

Aug 2023 - Present

Bangalore

- Architected a highly scalable, cloud-native compliance platform on OCI for enterprise Customer Screening (CS) and Transaction Filtering (TF). Leveraged OpenSearch and Oracle Text to process real-time and high-volume batch workloads, driven by a hybrid decision engine blending deterministic matching with AI semantic search.
- **Role Progression:** Promoted from Application Developer to Software Application Engineer (May 2026) for backend platform engineering, search architecture, performance optimization, and AI-enabled workflow design.
- **AI-Enabled Workflows:** Explored practical LLM-assisted workflows for enterprise operations and developer productivity while maintaining controlled, explainable, and reliable screening outcomes.

PROJECTS

• Matching & Scoring Engine

Java, Spring Boot, REST APIs, Oracle Text, OpenSearch, PL/SQL, OCI, Kubernetes, Microservices, Parallel Processing, Semantic Search, AI Similarity, LLM Models

- **Core Search Engine Migration:** Spearheaded migration from OpenSearch to Oracle Text, moving search-heavy matching closer to the database layer to improve scalability, reduce infrastructure cost, and simplify operational complexity.
- **Enterprise-Scale Screening:** Scaled screening workflows to process millions of transactions against billions of records, achieving 100+ TPS with sub-2.5 second average latency across 23,000-message volume runs and zero errors across 8 simultaneous global watchlists.
- **High-Volume Batch Processing:** Reduced 5K-transaction batch screening latency by 97%, from 2 hours 5 minutes to 3 minutes, using targeted database indexing, intelligent batching, aggregated query flows, and multi-threaded parallel processing with transaction isolation.
- **Real-Time Screening Optimization:** Reduced real-time screening latency by 85%, from 2 seconds to 300 milliseconds, by redesigning synchronous processing into an asynchronous pipeline with parallel database execution, thread management, and incremental result streaming.
- **Multi-Service Performance Tuning:** Improved throughput across Alert, Swift, and Data Transform services by moving heavy data processing from java layer into stored procedures, reducing database round trips by 70%, optimizing cache serialization, and adding strategic query caching.
- **Advanced Hybrid Scoring Engine:** Built a configurable scoring microservice combining deterministic matching rules, mathematical edit distances, fuzzy logic, semantic search, and AI similarity signals – including an LLM-assisted narrative text extraction workflow for unstructured entity matching – to reduce false positives and improve entity resolution accuracy for global watchlist screening.

• Enterprise RAG Knowledge Assistant

React, TypeScript, Spring Boot 3, WebFlux, Oracle 23ai, Vector Search, Vercel, Render

- **Full-Stack RAG Architecture:** Built and deployed an AI knowledge assistant using a React/TypeScript chat interface, Spring Boot WebFlux backend, Oracle 23ai vector-ready retrieval layer, and LLM integration.

- **Grounded Knowledge Ingestion:** Designed an admin-controlled ingestion pipeline for **7 curated knowledge sources**; chunks content into **2,200-character** segments with **220-character** overlap, enriches records with metadata, and prepares embedding-based retrieval.
- **Hybrid Retrieval & Reranking:** Implemented exact-text and semantic vector retrieval, returning the top **5** relevant evidence chunks per query; reranks results using vector similarity, source metadata, category, title, and keyword-intent signals.
- **Responsive Streaming Experience:** Added Server-Sent Events (SSE) for token streaming, session-aware chat history, response caching, a **24-second** request timeout, and source citations for every generated response.

EDUCATION

M.Tech in Information Technology, NIT Karnataka
B.Tech in Information Technology, JNTU Vizianagaram

Aug 2021 - July 2023 — CGPA: 8.96
 June 2016 - Sept 2020 — CGPA: 8.36

SKILLS

Languages	Java, SQL, PL/SQL
Frameworks & Architecture	Spring, Spring Boot, Hibernate, RESTful APIs, Microservices, Apache Kafka
Databases & Search	Oracle 19c, Oracle Text, OpenSearch, MySQL
AI, Search & Retrieval	Oracle 23ai Vector Search, Hybrid Search, Embeddings, RAG, Rule-Based Reranking, LLM Workflows, Prompt Design, Fuzzy Matching
Cloud & Tools	OCI, Grafana, APM, Git, Maven
Problem Solving	Data Structures & Algorithms, System Design (HLD/LLD) – 600+ problems solved (LeetCode)

CERTIFICATIONS & TRAINING

- **Oracle Cloud Infrastructure (OCI) AI Foundations** — *Oracle* (2024)
- **Oracle Database 19c: PL/SQL Workshop** — *Oracle* (2023)
- **Spring 6 with Spring Boot 3 & Java EE: Concurrency** — *LinkedIn Learning* (2024)
- **Advanced Core Java & Spring Boot Architecture** — *Udemy* (2026)
- **Generative AI with Large Language Models** — *DeepLearning.AI* (2026)
- **Introduction to Generative AI for Software Development** — *DeepLearning.AI* (2026)
- **Generative AI for Everyone** — *DeepLearning.AI* (2026)
- **Team Software Engineering with AI** — *DeepLearning.AI* (2026)
- **AI-Powered Software and System Design** — *DeepLearning.AI* (2026)
- **Model Context Protocol: Introduction and Advanced Topics** — *Anthropic* (2026)
- **Low-Level Design** — *takeUforward* (2026)

AWARDS & RECOGNITION

- **FSGIU Pacesetter – Spark Award (Oracle):** Awarded for achieving a 7x performance improvement by reducing latency from 5s to 700ms (86%) and playing a key role in migrating core search systems from OpenSearch to Oracle Text.
- **Oracle Agent IR Hackathon (2025):** Engineered an enterprise application to automate daily IUT check-in activities, securing 5th place out of 200 teams and earning official Oracle recognition for rapid prototyping and technical execution.

ACADEMIC ACHIEVEMENTS

- **Vice President of India Scholarship:** Awarded Scholarship for academic excellence (2019).
- **Competitive Examination Excellence:** Secured a **98.0 percentile** in GATE 2021 (CS/IT) and a top 2% statewide rank in EAMCET 2016, earning a full four-year merit tuition scholarship.